

Income and Crime Rate Association with Housing Prices Across California Counties

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Housing prices are an important topic in California because living costs are very high in many parts of the state. At the same time, California counties are not all the same. Some counties have much higher income levels, while others may have different public safety conditions. Because of these differences, it is interesting to study whether housing prices are associated with income and violent crime rates.

The main question of this project is:

What factors are associated with housing prices across California counties?

This project focuses on two smaller questions:

1. Do counties with higher median household income usually have higher housing prices?
2. Are counties with higher violent crime rates more likely to have lower housing prices?

These questions are meaningful because housing prices may reflect more than only the size or location of homes. They may also be related to the economic situation of residents and to the general conditions of a county. Looking at income and violent crime together with housing prices can provide a broader understanding of county level differences across California.

To study this topic, the project combines three types of data: county level housing values, median household income, and violent crime rates. After the datasets are cleaned and joined, they can be compared in one integrated table. The analysis uses descriptive statistics and visualizations, including scatter plots, bar charts, and comparison graphs, to identify visible patterns in the data.

This project is exploratory. It does not try to prove that income or crime directly causes changes in housing prices. Instead, it examines whether these variables appear to be associated with one another across California counties.

The source links are listed as:

1. Zillow research housing data: county level housing value dataset downloaded as csv file
<https://www.zillow.com/research/data>
2. U.S. Census Bureau American Community Survey (ACS): county level median household income collected through an API
<https://www.census.gov/data.html>
3. California Department of Justice Crime Data: county level violent crime data available through an online data portal
<https://oag.ca.gov/crime>

These data sources will be combined mainly by **county name** and **state**. After collecting the data, I will clean the files and standardize county names, so they match across all sources. Then I will create one integrated county level dataset that includes housing prices, median household income and violent crime rates for each county. This combined dataset will make it possible to compare counties directly and analyze whether these variables are related to housing prices. Some additional cleaning will be necessary because the sources may use different file formats, variable names, or county labels.